

PROJECT DOCUMENTATION

Leave Management System

An Automated Web-Based Solution for Employee Leave Management

Project Title	Leave Management System
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1. Project Team Members

The following students collaborated on the design, development, and documentation of the Leave Management System project:

S. No.	Name	Roll No.	Role
1	Varun Kushwah	2400290120275	Team Lead / Backend Development
2	Vaibhav Gupta	2400290120270	Frontend Development
3	Tanya Bhadana	2400290120251	Database Design & Testing
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2. Problem Background

Leave management is a routine yet critical administrative process carried out in every organization, institution, and academic department. Traditionally, this process is handled manually — through paper-based leave application forms, physical registers, and in-person approvals from supervisors or HR personnel. Employees or staff members are required to fill out a leave request form, physically submit it to the concerned authority, and wait for manual verification and approval, which is then recorded in a register or spreadsheet.

As organizations grow in size and complexity, this manual approach becomes increasingly inefficient. Records get misplaced, approvals are delayed due to the unavailability of the approving authority, and there is no centralized system to track leave balances, leave history, or department-wise leave statistics. HR departments spend a significant amount of time manually consolidating leave data, which is prone to human error and lacks transparency for employees who wish to check their leave status in real time.

3. Problem Statement

There is a need for a centralized, digital Leave Management System that allows employees to apply for leave online, enables managers or administrators to review and approve or reject requests efficiently, and automatically maintains accurate, real-time records of leave balances and leave history — thereby eliminating the delays, errors, and lack of transparency associated with manual, paper-based leave processing.

4. Impact

The absence of an automated leave management solution has several tangible negative effects on an organization:

- Delayed approvals due to dependency on the physical presence of the approving authority.
- Loss or mismanagement of physical leave records and application forms.
- Inaccurate tracking of leave balances, leading to disputes between employees and management.
- Increased administrative workload for HR staff who must manually compile leave data and generate reports.
- Lack of transparency, as employees have no direct visibility into the status of their leave requests.
- Inefficient resource and workforce planning, since managers cannot easily view team-wide leave schedules.

By addressing these issues, an automated system directly improves organizational efficiency, employee satisfaction, and the accuracy of administrative records.

5. Proposed Software Solution

The proposed Leave Management System (LMS) is a web-based application designed to digitize and automate the entire leave request and approval workflow. The system provides role-based access for Employees, Managers/Approvers, and Administrators, and includes the following core features:

- Online leave application submission with leave type selection (casual, sick, earned, etc.) and supporting document upload.
- Automated routing of leave requests to the appropriate approving authority.
- Real-time approval or rejection workflow with email/in-app notifications.
- Dynamic tracking of leave balances, accruals, and leave history for each employee.
- An administrative dashboard for HR to configure leave policies, holidays, and generate analytical reports.
- A calendar view to visualize team and department-wide leave schedules.
- Secure authentication and role-based access control to protect sensitive employee data.

The system is built using a modern web technology stack, with a relational database to ensure data integrity, and a responsive front end so that the application is accessible across desktop and mobile devices.

6. Technical Feasibility

The Leave Management System is technically feasible to develop and deploy within the scope of this project, based on the following considerations:

6.1 Technology Availability

- The proposed technology stack (front-end framework, back-end framework/API layer, and relational database) consists of mature, well-documented, and freely available open-source technologies.
- Standard libraries and frameworks are available for authentication, role-based access control, and notification handling, reducing development risk.

6.2 Team Skill Set

- The project team possesses the requisite academic knowledge of web development, database design, and software engineering principles required to implement the system.
- The modular nature of the system (authentication, leave workflow, reporting) allows tasks to be distributed efficiently among team members.

6.3 Infrastructure Requirements

- The system can be developed and tested on standard personal computers/laptops without specialized hardware.
- Deployment is possible on low-cost or free-tier cloud hosting platforms, making the solution practical within an academic project's resource constraints.

7. Operational Feasibility

The system is operationally feasible and expected to integrate smoothly into existing organizational workflows for the following reasons:

7.1 Ease of Use

- The system replicates and streamlines a process users are already familiar with (submitting and approving leave), minimizing the learning curve.
- An intuitive, form-based interface ensures that employees and managers with minimal technical expertise can use the system effectively.

7.2 Organizational Acceptance

- The system reduces the manual workload of HR personnel and approving managers, making it likely to be well-received by end users.

- Automated notifications and status tracking directly address existing pain points, increasing the likelihood of user adoption.

7.3 Maintainability

- A modular system architecture ensures that future enhancements (e.g., new leave types or approval workflows) can be added with minimal disruption.
- Centralized digital records simplify auditing, reporting, and long-term data maintenance compared to paper-based systems.

8. Project Scope

8.1 In-Scope

Included in this Project
▪ Employee registration, login, and role-based authentication (Employee, Manager, Administrator).
▪ Online submission of leave applications with leave type, date range, and reason/document upload.
▪ Approval / rejection workflow routed to the designated approving authority.
▪ Real-time leave balance calculation and leave history tracking for each employee.
▪ Email or in-app notifications for application status updates.
▪ Administrative dashboard for managing leave policies, public holidays, and department settings.
▪ Calendar view of approved leaves for teams and departments.
▪ Basic reporting and analytics (e.g., leave trends, department-wise summaries) for administrators.

8.2 Out-of-Scope

Excluded from this Project
▪ Integration with third-party payroll or salary-processing systems.
▪ Biometric or physical attendance tracking hardware integration.
▪ Multi-organization / multi-tenant support (system is designed for a single organization instance).
▪ Native mobile applications (the system will be delivered as a responsive web application only).
▪ Advanced AI-based leave prediction or workforce optimization features.
▪ Integration with external HRMS (Human Resource Management System) platforms.

9. Expected Outcome

Upon completion, the Leave Management System is expected to deliver the following outcomes:

- A fully functional, web-based platform that digitizes the complete leave application and approval lifecycle.
- Significant reduction in the time taken to process and approve leave requests, compared to manual methods.
- Improved accuracy and transparency in leave balance tracking, reducing disputes and administrative errors.
- A centralized repository of leave data that supports easy reporting, auditing, and organizational planning.
- Enhanced employee experience through self-service access to leave status and history.
- A scalable and maintainable system architecture that can be extended with additional features in the future.

Overall, the project demonstrates the practical application of software engineering, database management, and web development concepts to solve a real-world organizational problem, while delivering a solution of tangible, measurable value.